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(1)

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Question (2) =>

$x = S_x v + t_x$

$y = S_y w + t_y$

$A(3,3) \rightarrow A'(6.5, 3.4) \rightarrow$ $6.5 = S_x * 3 + t_x \rightarrow (1)$

$3.4 = 3 S_y + t_y \rightarrow (2)$

$B(7,3) \rightarrow B'(12.5, 3.4) \rightarrow$ $12.5 = 7 S_x + t_x \rightarrow (3)$

$3.4 = 3 S_y + t_y \rightarrow (4)$

$C(7,7) \rightarrow C'(12.5, 6.6) \rightarrow$ $12.5 = 7 S_x + t_x \rightarrow (5)$

$6.6 = 7 S_y + t_y \rightarrow (6)$

From (1), (3) $\rightarrow$ $S_x = 1.5$, $t_x = 2$

From (2), (6) $\rightarrow$ $S_y = 0.8$, $t_y = 1$

Scaling = $\begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$, Translation = $\begin{bmatrix} 0 & 0 & t_x \\ 0 & 0 & t_y \\ 0 & 0 & 1 \end{bmatrix}$

Transformation matrix = $\begin{bmatrix} 1.5 & 0 & 2 \\ 0 & 0.8 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

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if rounding S_x, S_y => $\begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Question (3) => Form GA-TA-H1

Graph D - TA - H2

↳ image is brightened a little bit ⇝ power law with $\gamma < 1$

↳ $\gamma \approx 1$

Graph E - TB - H3

↳ negative transformation → $s = (L-1) - r$

Graph B - TC - H4

↳ all the image is darkened → power law (gamma) with $\gamma > 1$

Graph C - TD - H1

↳ all the image is brightened → power law with $\gamma < 1$

Graph F - TE - H5

↳ Thresholding, intensities below and above a certain threshold only.

Graph A is not used.

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Question (4) =>

$M \times N = 32$, $L = 8$ levels

Histogram eq. For input image

(3)

r_k	n_k	$P(r_k)$	CDF	$\rightarrow$ CDF * (L-1)	S_k Equalized levels	$P_s(S_k)$
0	1	1/32	1/32	0.218	$\rightarrow$ 0	1/32
1	4	1/8	5/32	1.093	1	1/8
2	8	1/4	13/32	2.84	3	1/4
3	5	5/32	9/16	3.93	4	$\rightarrow$ 7/32
4	2	1/16	5/8	4.37	4	
5	2	1/16	11/16	4.8	5	1/16
6	5	5/32	27/32	5.9	6	5/32
7	5	5/32	1	7	7	5/32

z_q	n_q	$P_z(z_q)$	CDF	CDF * (L-1)	
0	0	0	0	0	0
1	0	0	0	0	0
2	4	1/8	1/8	0.87	1
3	4	1/8	1/4	1.7	2
4	4	1/8	3/8	2.6	3
5	4	1/8	1/2	3.5	4
6	5	5/32	21/32	4.59	5
7	11	11/32	1	7	7

	Hist. eq of specified		Hist. eq of input	
0 →	0	→	0	1/32
1 →	0		1	1/8
2 →	1		3	1/4
3 →	2		4	] → 7/32
4 →	3		4	
5 →	4		5	1/16
6 →	5		6	5/32
7	7	→	7	5/32

$\frac{3.5}{5}$

Grey Levels	0	1	2	3	4	5	6	7
Probabilities	$\frac{1}{32}$	$\frac{1}{32}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{7}{32}$	$\frac{1}{16}$	$\frac{5}{32}$

Image grid ⇒

	1	2	3	4	5	6	7	8
1	7	6	6	0	2	2	2	2 ✓
2	7	6	5	2	2	2	4	7 ✓
3	6	6	4	2	2	2	4	7 ✓
4	6	6	4	2	4	2	5	7 ✓

-0.5

Not correct!

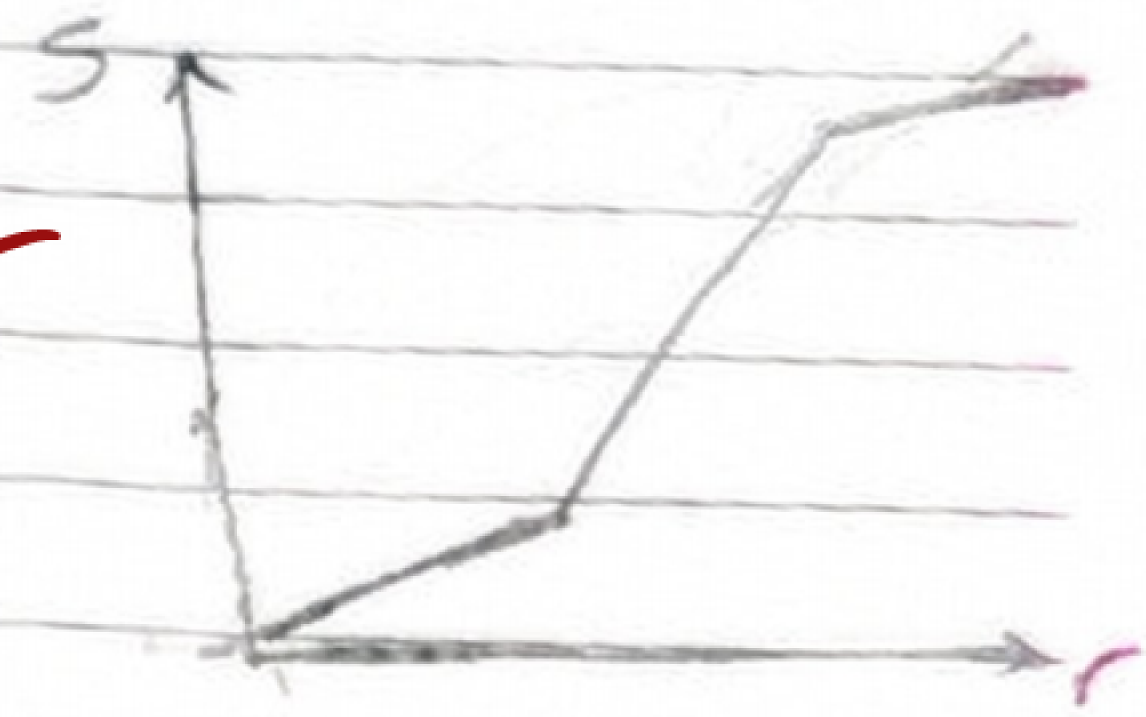
(*) Not ~~accurate~~ (no time)

mapping isn't clear! (-1)

Intensity values => 0, 50, 100, 150, 200, 255

For enhancing this image =>

2- will make a contrast stretching transformation to enhance the image due to the darkened parts.



3- Then will apply local histogram equalization for the parts which still in need for that.

1- a median filter is used to reduce the noise

4- a laplacian filter is used to enhance image details.

(*) look at the numbering of steps plz.

-0.5 Binary ممتاز بس اسمه مش

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-0.5 Histogram مفيش